

Han Yang

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Education

Peking University , Materials Science and Engineering

2023.09 - Present

GPA: 3.533/4.0 , 114 credits completed to date

Course: Mathematical Analysis, Linear Algebra, Fundamentals of Materials Science, Materials Chemistry, Materials Physics, Computational Materials Science and Engineering, Materials Science and Engineering Experiment, Programming in AI (**Grades 98**), The Computational Foundations of Neural Networks (**Grades 98**)

Research Interests

Machine Learning, Computational Materials Science, Electrocatalyst Design

Experience

Zou Lab Undergraduate Lab Member

2024.09 - Present

Developing hands-on experience in materials synthesis, sample preparation, laboratory operations, and characterization workflows, including SEM, XPS, XRD, and related analytical instruments.

Zou Lab Undergraduate Research Project

2025.09 - Present

Studying support design for dual-atom catalysts in the oxygen reduction reaction (ORR), with a focus on synthesizing support materials that stabilize paired active sites and relating support structure to electrocatalytic performance.

The Computational Foundations of Neural Networks

Course Project

Reproduced [CEBRA](#), a PyTorch self-supervised contrastive learning framework for joint behavioral and neural time-series analysis; rebuilt data sampling, model training, latent-embedding visualization, and decoding-oriented evaluation following the [Nature paper](#).

Selected Projects

Clab V2

2025.05 - 2025.11

Systems Team Member in the Clab V2 Development Group. CLab is an LCPU-built open cloud computing platform for PKU courses, providing ready-to-use programming environments for 15+ courses and 1,000+ users across 43 campus units; participated in the V2 systems effort to improve platform stability, performance, high availability, and user experience for HPC-oriented workloads.

PKUHUB Core Developer and Operations Lead

2025.04 - Present

Served as a core developer of PKUHUB, leading core refactoring and operations for a campus note-taking platform with approximately 5K users, 1K+ shared resources, and 30K+ downloads.

Honors & Awards

Tongcheng New Materials Scholarship

2023.09-2024.07

Skills

English:

Programming:

Scientific Computing:

Tools:

CET-4, CET-6

Python, C/C++, Rust, Typst, LaTeX, Node.js

NumPy, pandas, Matplotlib, scikit-learn, PyTorch, Jupyter

Git, Linux, Docker, Conda